

E-commerce Customer Journey & Conversion Funnel Analytics Report

SQL Server | Power BI | DAX | Python-ready analytical workflow

This report documents an end-to-end Data Analyst portfolio project analysing an e-commerce customer journey from session engagement to completed purchase. The dashboard identifies high-performing channels, valuable customer segments, funnel drop-off stages, and actionable opportunities to improve conversion performance.

Business Question

Which traffic sources, devices, and customer segments drive the highest conversion, and where do users drop off before purchase?

Item	Description
Domain	Marketing Analytics
Primary tools	SQL Server, Power BI, DAX
Analytical extension	Python for session-level funnel progression and EDA
Dashboard pages	Executive Overview, Channel Performance, Customer Segments, Conversion Opportunities
Main grain	One row per session in fact_session_funnel

The project follows a business-first analytics workflow. The analysis starts from a business question, moves through SQL cleaning and model design, defines a reusable metric layer, and ends with Power BI reporting and action-focused recommendations.

- Business Question: define the marketing and conversion problem.
- SQL Cleaning: clean and validate raw customer, session, interaction, and purchase data.
- Metric Layer: define traffic, revenue, conversion, funnel transition, and drop-off measures.
- Power BI Dashboard: build interactive pages for executive, channel, customer, and opportunity analysis.
- Python Analysis: extend the analysis with session-level funnel progression and drop-off validation.
- Recommendation: translate funnel findings into stage-based business actions.

Raw e-commerce CSV files were imported into SQL Server as flat tables. SQL Server was used to profile, clean, validate, and transform the data before loading it into Power BI. The final reporting model focused on customer, session, interaction, and purchase tables because these directly support funnel analysis.

- Checked row counts to confirm successful import of all source tables.
- Checked missing values across key fields such as user_id, session_id, interaction_id, purchase_id, start_time, device_type, referrer_source, interaction_type, quantity, unit_price, and total_amount.
- Checked duplicate keys for user_id, session_id, interaction_id, and purchase_id.
- Validated invalid values such as negative dwell time, invalid age, invalid quantity, and negative transaction values.
- Recalculated purchase amount using quantity x unit_price and compared it with total_amount to validate transaction accuracy.

- Checked relationship integrity across users, sessions, interactions, and purchases.
- Created clean views and transformed them into final reporting tables for Power BI.

The Power BI model uses a simplified star schema with one main fact table and supporting dimensions. The main fact table is fact_session_funnel, which has one row per session. This enables the dashboard to analyse each session by funnel stage, drop-off stage, channel, device, customer segment, and date.

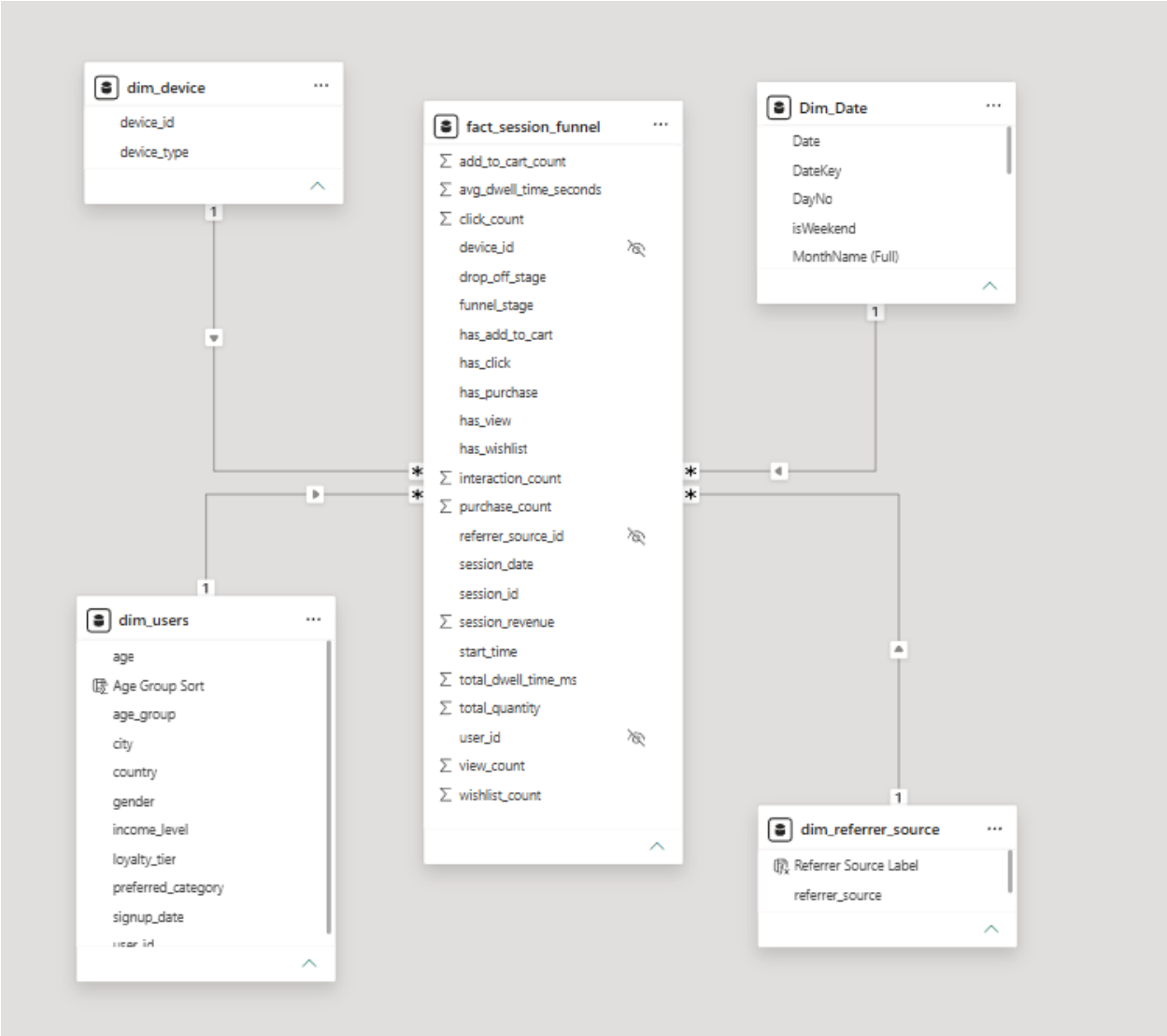


Figure 1. Power BI model schema showing the session-level star schema.

Item	Description
fact_session_funnel	Session-level fact table containing funnel flags, interaction counts, purchase counts, session revenue, funnel stage, and drop-off stage.
dim_users	Customer attributes such as age, age group, gender, country, city, income level, preferred category, and loyalty tier.
dim_device	Device type dimension used to analyse conversion by desktop, mobile, or tablet.
dim_referrer_source	Traffic source dimension used to compare source-level conversion and drop-off performance.
Dim_Date	Date table created in Power BI to support time-based analysis and Australian date formatting.

The model relationships are one-to-many from each dimension to fact_session_funnel, with single filter direction. This avoids unnecessary many-to-many relationships and supports reliable cross-filtering across dashboard pages.

A reusable DAX metric layer was created to standardise calculations across the report. Measures were organised into dedicated measure tables for clarity, including core measures, funnel step measures, conversion rate measures, funnel transition measures, drop-off measures, interaction measures, revenue quality measures, and time intelligence measures.

Measure	Purpose	DAX / Logic
Total Sessions	Counts unique sessions in the selected filter context.	DISTINCTCOUNT(fact_session_funnel[session_id])
Total Users	Counts unique users in the selected filter context.	DISTINCTCOUNT(fact_session_funnel[user_id])
Total Revenue	Calculates total session revenue.	SUM(fact_session_funnel[session_revenue])
Total Purchases	Calculates total purchase count.	SUM(fact_session_funnel[purchase_count])
Average Order Value	Average revenue per purchase.	DIVIDE([Total Revenue], [Total Purchases])
Revenue per Session	Average revenue generated per session.	DIVIDE([Total Revenue], [Total Sessions])
Revenue per User	Average revenue generated per user.	DIVIDE([Total Revenue], [Total Users])

Measure	Purpose	DAX / Logic
Sessions with View	Sessions that reached the view stage.	CALCULATE([Total Sessions], fact_session_funnel[has_view] = TRUE())
Sessions with Click	Sessions that reached the click stage.	CALCULATE([Total Sessions], fact_session_funnel[has_click] = TRUE())
Sessions with Add to Cart	Sessions that reached the add-to-cart stage.	CALCULATE([Total Sessions], fact_session_funnel[has_add_to_cart] = TRUE())
Converted Sessions	Sessions that completed purchase.	CALCULATE([Total Sessions], fact_session_funnel[has_purchase] = TRUE())
Purchase Conversion Rate	Percentage of sessions that converted to purchase.	DIVIDE([Converted Sessions], [Total Sessions])
View to Click Rate	Percentage of viewed sessions that clicked.	DIVIDE([Sessions with Click], [Sessions with View])
Click to Cart Rate	Percentage of clicked sessions that added to cart.	DIVIDE([Sessions with Add to Cart], [Sessions with Click])
Cart to Purchase Rate	Percentage of cart sessions that completed purchase.	DIVIDE([Converted Sessions], [Sessions with Add to Cart])

Measure	Purpose	DAX / Logic
Drop-off Share	Share of sessions within each selected drop-off stage.	DIVIDE([Total Sessions], CALCULATE([Total Sessions], ALLSELECTED(fact_session_funnel[drop_off_stage]))))
Non-converted Sessions	Sessions that did not complete purchase.	[Total Sessions] - [Converted Sessions]
Non-conversion Rate	Share of sessions that did not purchase.	DIVIDE([Non-converted Sessions], [Total Sessions])
Cart Drop-off Sessions	Sessions that dropped after cart.	CALCULATE([Total Sessions], fact_session_funnel[drop_off_stage] = "Drop after Cart")

The Executive Overview page provides a high-level summary of overall funnel health. It shows traffic, revenue, conversion rate, average order value, revenue per session, the customer journey funnel, monthly conversion trend, sessions by drop-off stage, and conversion rate by traffic source.

- Main question: How is the overall funnel performing?
- The dashboard shows whether current traffic and revenue are healthy.
- It identifies the current purchase conversion rate and where users drop off before purchase.
- It compares conversion rate by traffic source to identify stronger-performing sources.

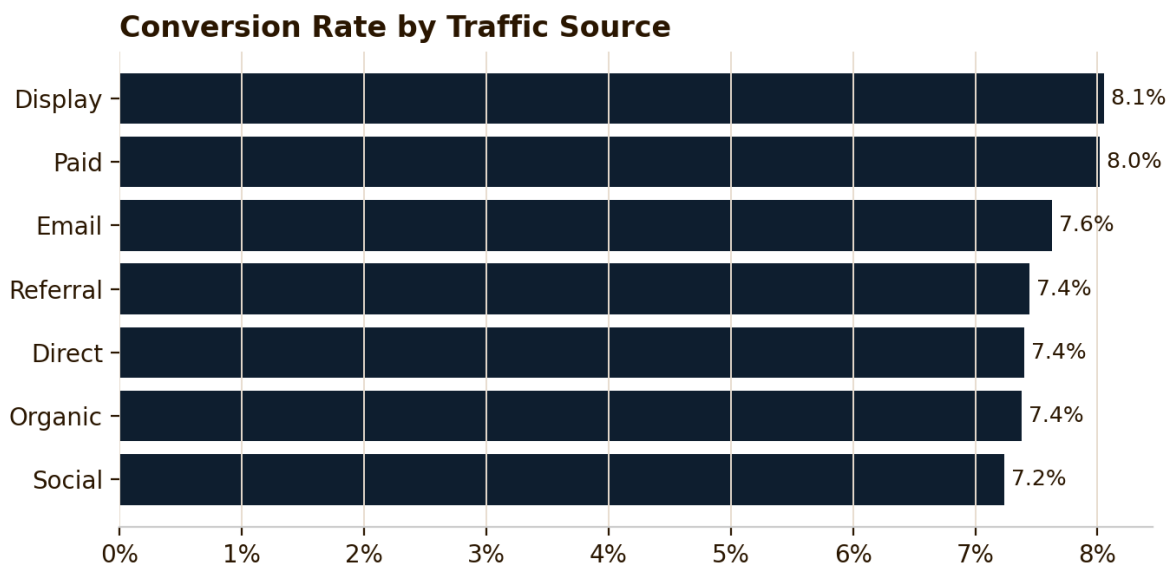


Figure 2. Conversion rate by traffic source.

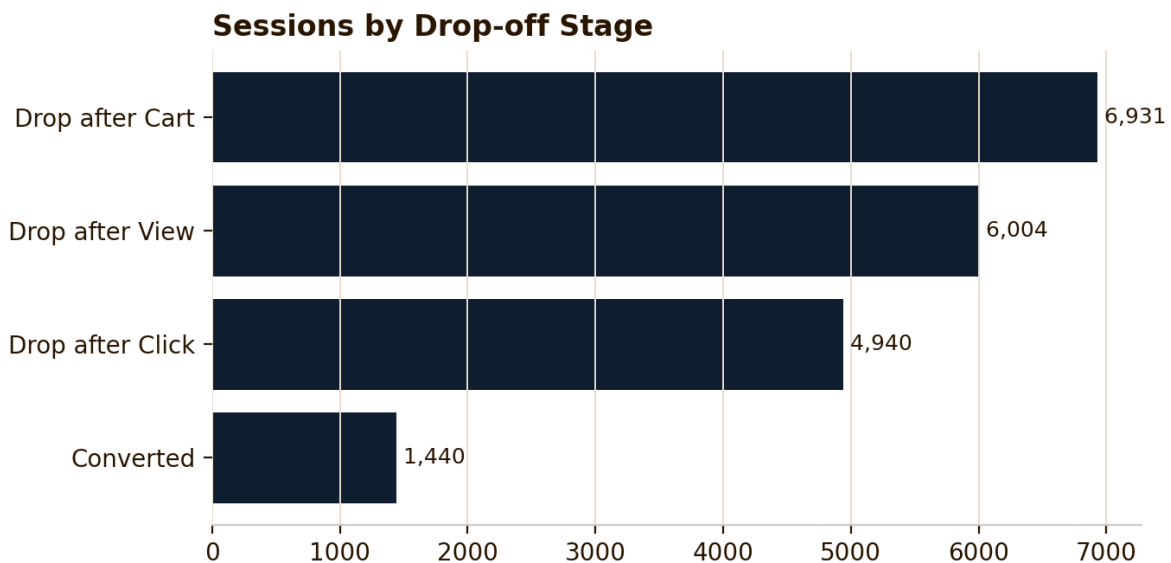


Figure 3. Sessions by drop-off stage.

The Channel Performance page evaluates how traffic sources and device types contribute to conversion and revenue quality. It compares session volume, conversion rate, revenue per session, and source-device performance.

- Main question: Which traffic sources and devices drive better conversion and revenue quality?
- The page highlights that traffic volume alone does not fully explain channel success.
- The scatter plot identifies high-traffic/high-conversion channels, low-traffic/high-conversion opportunities, and high-traffic/low-conversion optimisation areas.

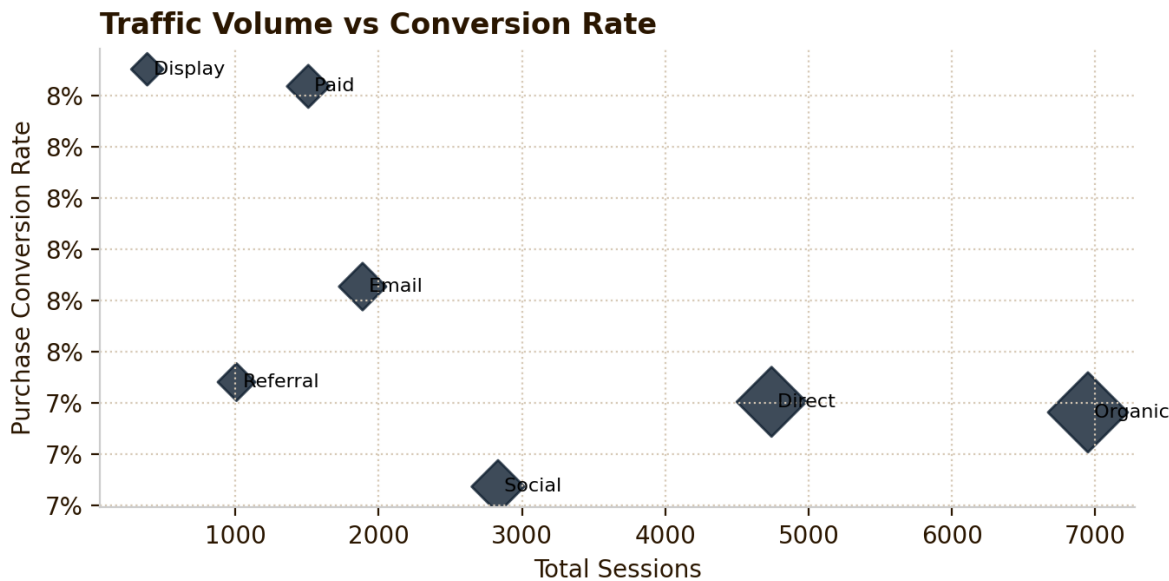


Figure 4. Traffic volume versus conversion rate by source.

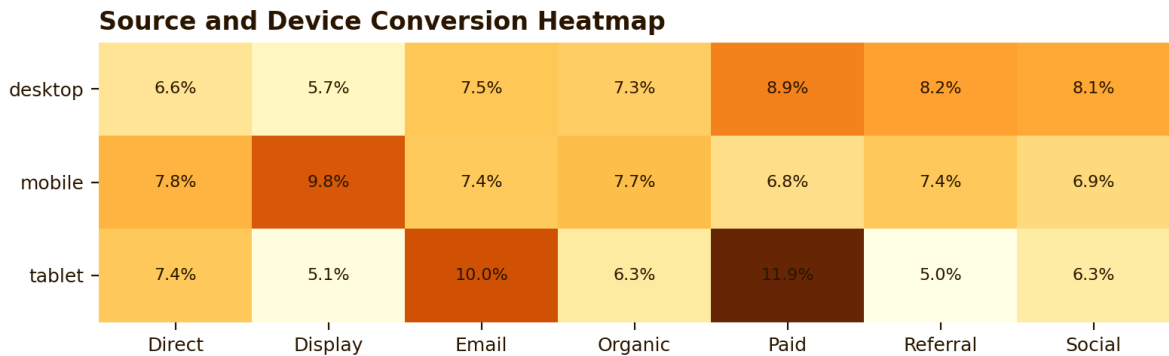


Figure 5. Source and device conversion heatmap.

The Customer Segment Analysis page identifies customer groups that generate stronger revenue per session and conversion. It analyses loyalty tier, income level, age group, traffic source, and device type to support more targeted marketing decisions.

- Main question: Which customer segments should the marketing team prioritise for targeting?
- The decomposition tree breaks down revenue per session by loyalty tier, income level, and age group.
- The age group chart compares session volume and conversion rate across customer age groups.
- The heatmap shows which source-device combinations support better conversion.

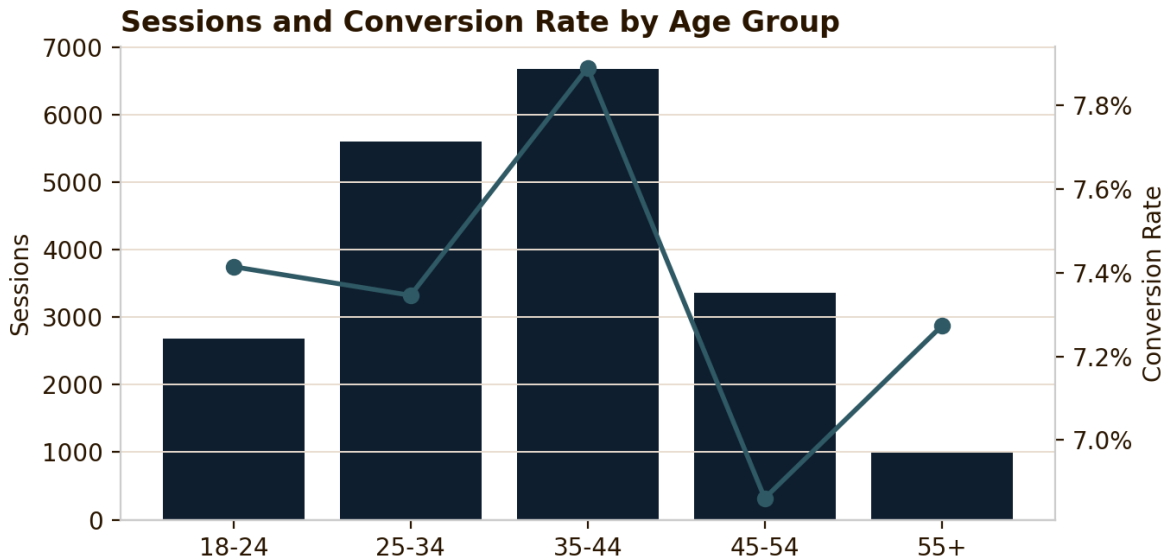


Figure 6. Sessions and conversion rate by age group.

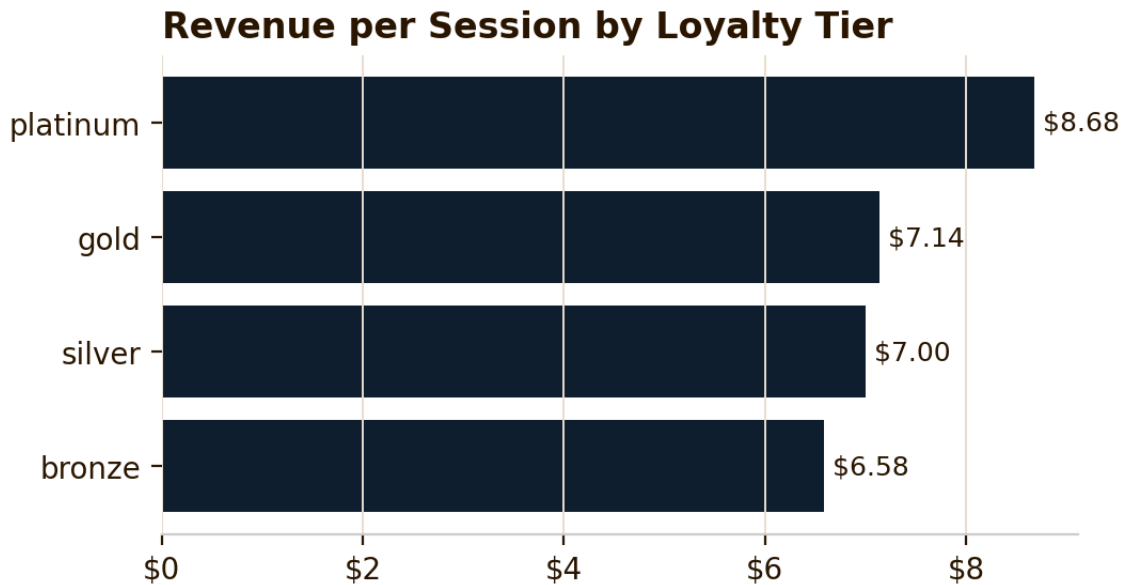


Figure 7. Revenue per session by loyalty tier.

The Conversion Opportunities page turns the dashboard findings into business actions. It focuses on the weakest funnel transition, largest drop-off stage, source-level drop-off patterns, and a stage-based action guide.

- Main question: Where are the biggest conversion opportunities, and what should the business do next?
- The weakest funnel transition is the final Cart -> Purchase step.
- Drop after Cart is the largest drop-off stage, indicating users show intent but do not complete purchase.
- The source-level heatmap identifies which traffic sources contribute most to each drop-off stage.

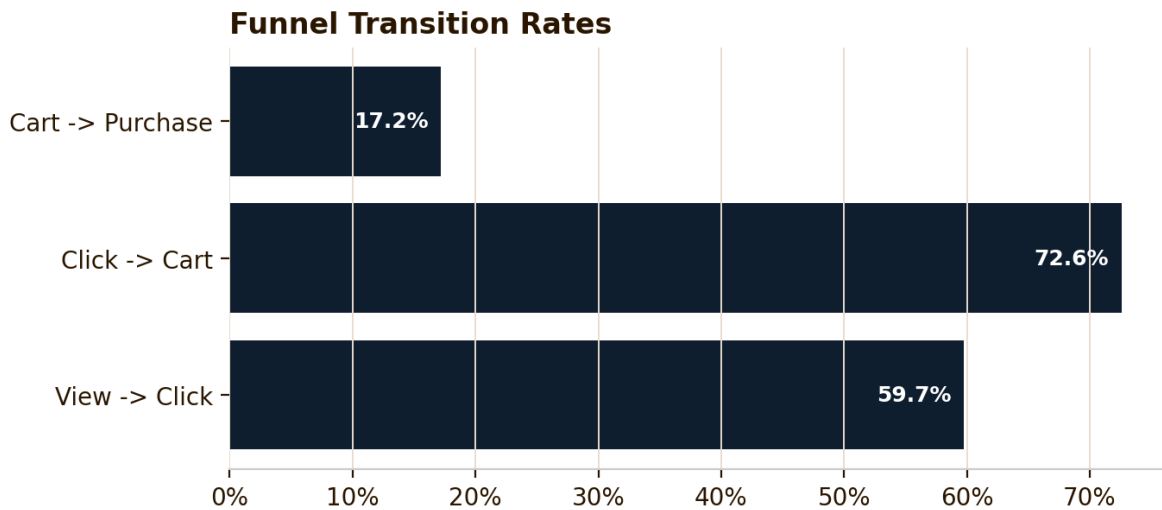


Figure 8. Funnel transition rates.

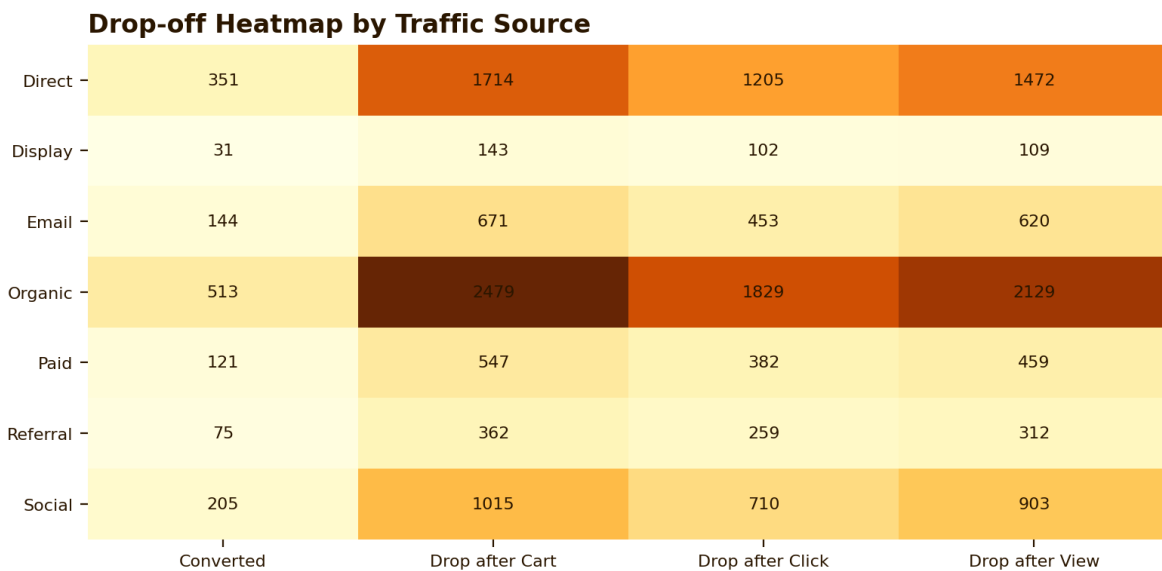


Figure 9. Drop-off heatmap by traffic source.

1. Cart abandonment is the largest conversion opportunity. The weakest funnel step is Cart -> Purchase, and the largest drop-off stage is Drop after Cart. This indicates that users are showing purchase intent but facing friction before completing checkout.
2. Traffic volume does not always equal conversion quality. Some traffic sources generate high session volume but do not necessarily produce the highest conversion rate or revenue per session.
3. Customer value differs across segments. Revenue per session varies by loyalty tier, income level, and age group, suggesting that marketing campaigns should be tailored to high-value segments.
4. Source and device combinations matter. Conversion rates differ across referrer source and device type, meaning campaign optimisation should consider both channel and device behaviour.
5. Recommendations should depend on the drop-off stage. A user dropping after view requires a different intervention compared with a user dropping after cart.

The recommendations use a stage-based action framework. This makes the report flexible: when dashboard filters change, the user can still match the largest drop-off stage or weakest transition with the most relevant action.

- Send cart reminder emails or remarketing ads to users who dropped after cart.
 - Review checkout speed, form length, payment options, and account creation requirements.
 - Improve transparency around delivery fees, return policy, and total price.
 - Test limited-time offers, free shipping thresholds, or checkout incentives for high-intent users.
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- Improve landing page relevance based on the selected traffic source.
 - Make product value propositions clearer above the fold.
 - Improve product images, descriptions, ratings, and trust signals.
 - Strengthen call-to-action placement and wording.
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- Prioritise high-converting channels with strong revenue per session.
 - Review high-traffic but low-conversion channels for landing page or audience mismatch.
 - Use the source-level heatmap to identify which channels create the most drop-off.
 - Avoid reducing campaign spend based only on low conversion rate without checking revenue per session.
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- Prioritise segments with higher revenue per session and conversion rate.
 - Personalise offers by loyalty tier, income level, and age group.
 - Use customer value drivers to guide campaign targeting.
 - Compare customer segment performance with source-device behaviour before launching campaigns.

Item	Description
Drop after View	Improve landing page relevance, product content, and CTA visibility.
Drop after Click	Review product pages, pricing clarity, trust signals, and recommendations.
Drop after Cart	Use cart recovery campaigns and optimise checkout friction.
Weak Cart -> Purchase	Investigate checkout barriers such as hidden fees, slow checkout, or limited payment options.
High source-level drop-off	Use the heatmap to prioritise traffic sources with the highest drop-off volume.

This dashboard helps the business move from descriptive reporting to decision-focused analytics. Instead of only showing what happened, it identifies where users leave the funnel and what actions can be taken.

- Improved conversion rate through checkout optimisation.
- Better campaign targeting through customer segment analysis.
- More efficient marketing spend allocation by traffic source.
- Higher revenue per session through high-value segment targeting.
- Reduced cart abandonment through retargeting and checkout improvements.

This project demonstrates an end-to-end Data Analyst workflow, starting from a business question and ending with actionable recommendations. SQL Server was used to clean and validate the raw data, Power BI was used to build a structured star schema and dashboard, and DAX measures were created to track traffic, conversion, revenue, and drop-off performance.

The key finding is that the largest opportunity sits near the end of the funnel, where users add items to cart but do not complete purchase. This indicates that the business should focus on cart recovery, checkout optimisation, and source-level drop-off investigation.

Overall, the dashboard provides a clear framework for improving e-commerce conversion performance by combining funnel analysis, channel performance, customer segmentation, and action-based recommendations.